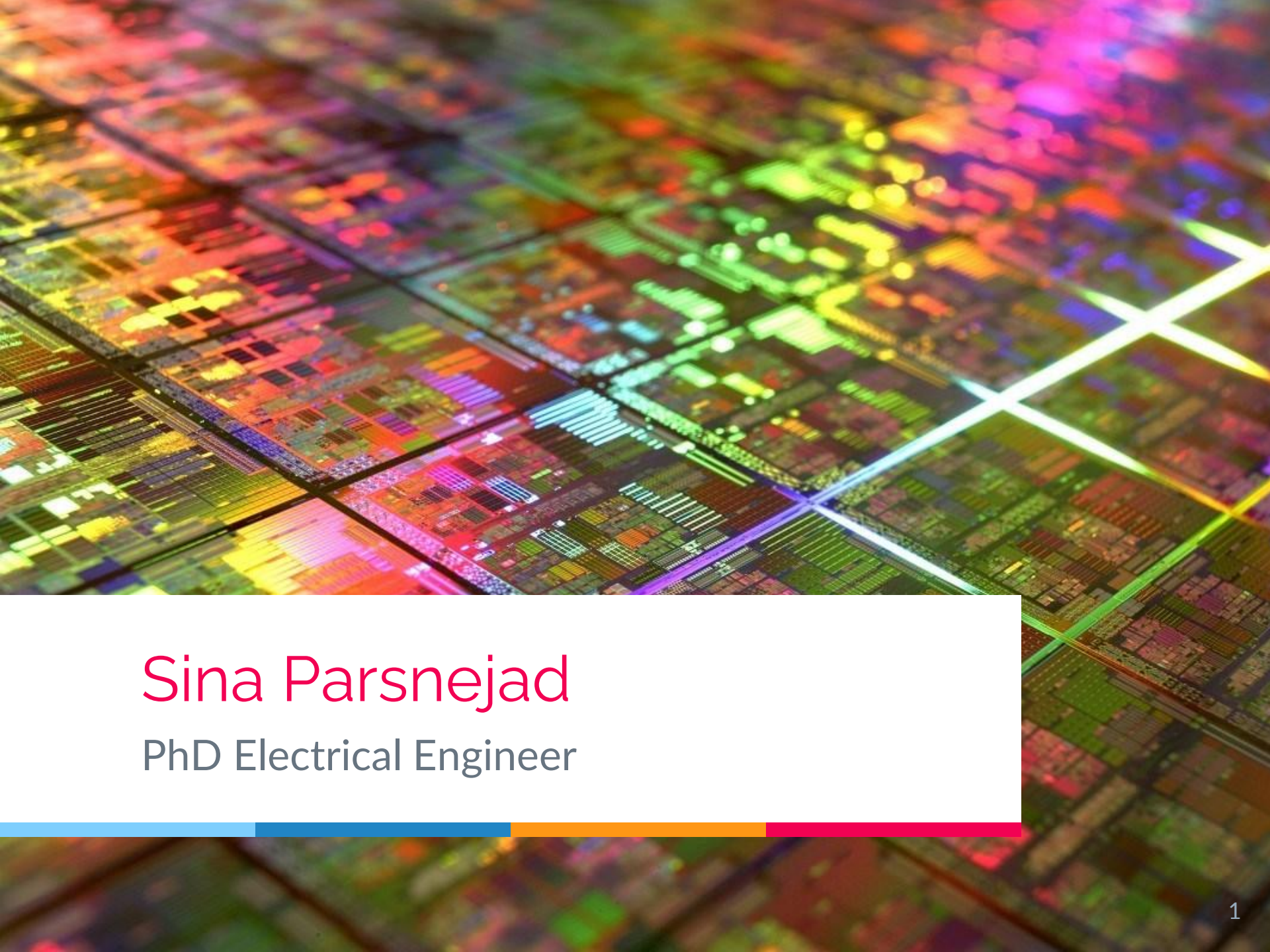




Sina Parsnejad

PhD Electrical Engineer

Hello!

I am passionate about finding solutions for problems in unconventional fields of electronics and conducting research on leading edge technologies.

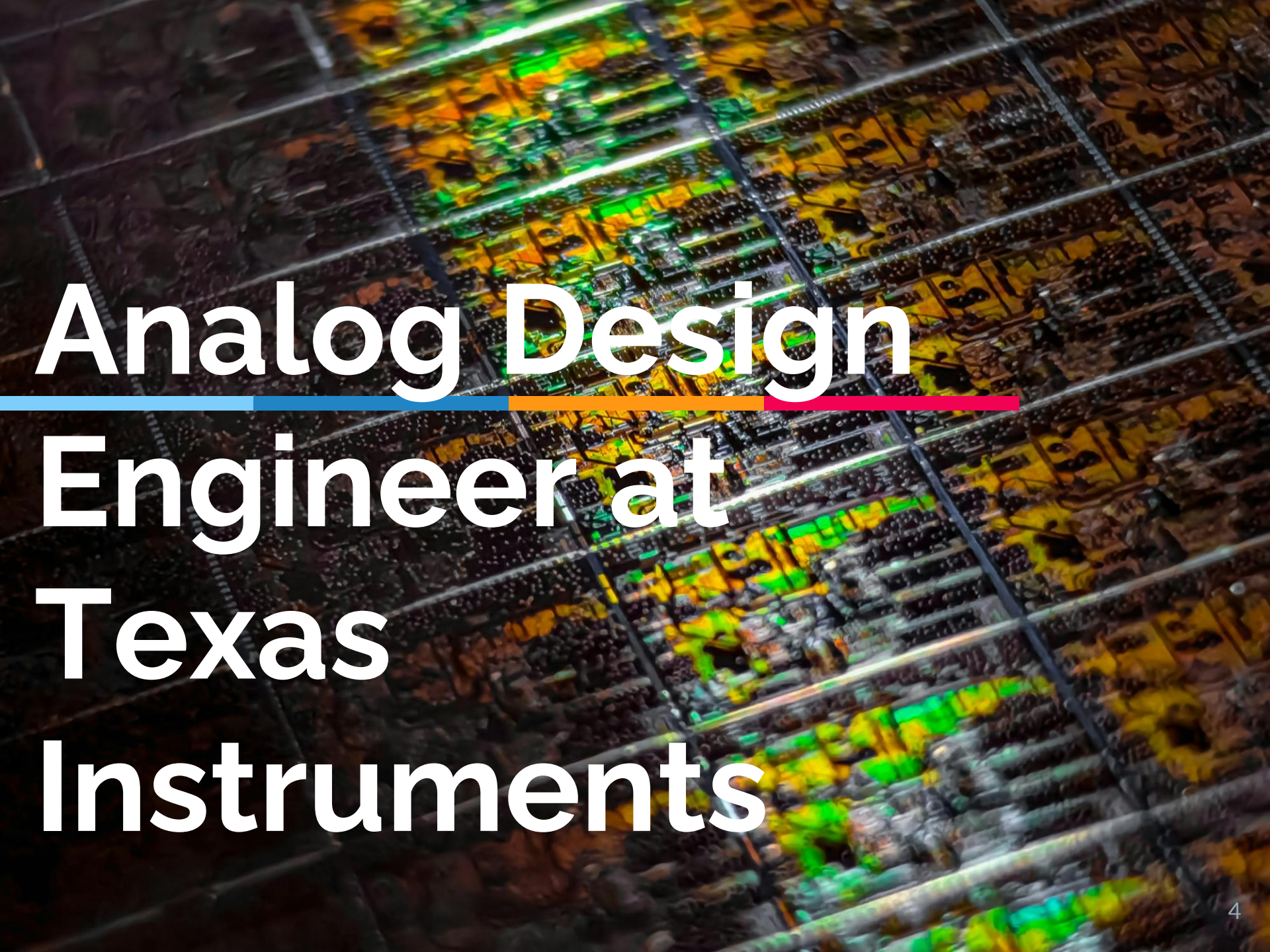
Sina Parsnejad
2/10/2026



A close-up photograph of a circular CMOS chip mounted on a wafer. The chip's surface is covered with a rainbow-colored grid pattern, likely representing a thermal or electrical map. The wafer is held in a metal chuck, and a warning label is visible in the bottom left corner.

CMOS designs

Over the years, I have been involved in multiple Analog and mix signal design projects



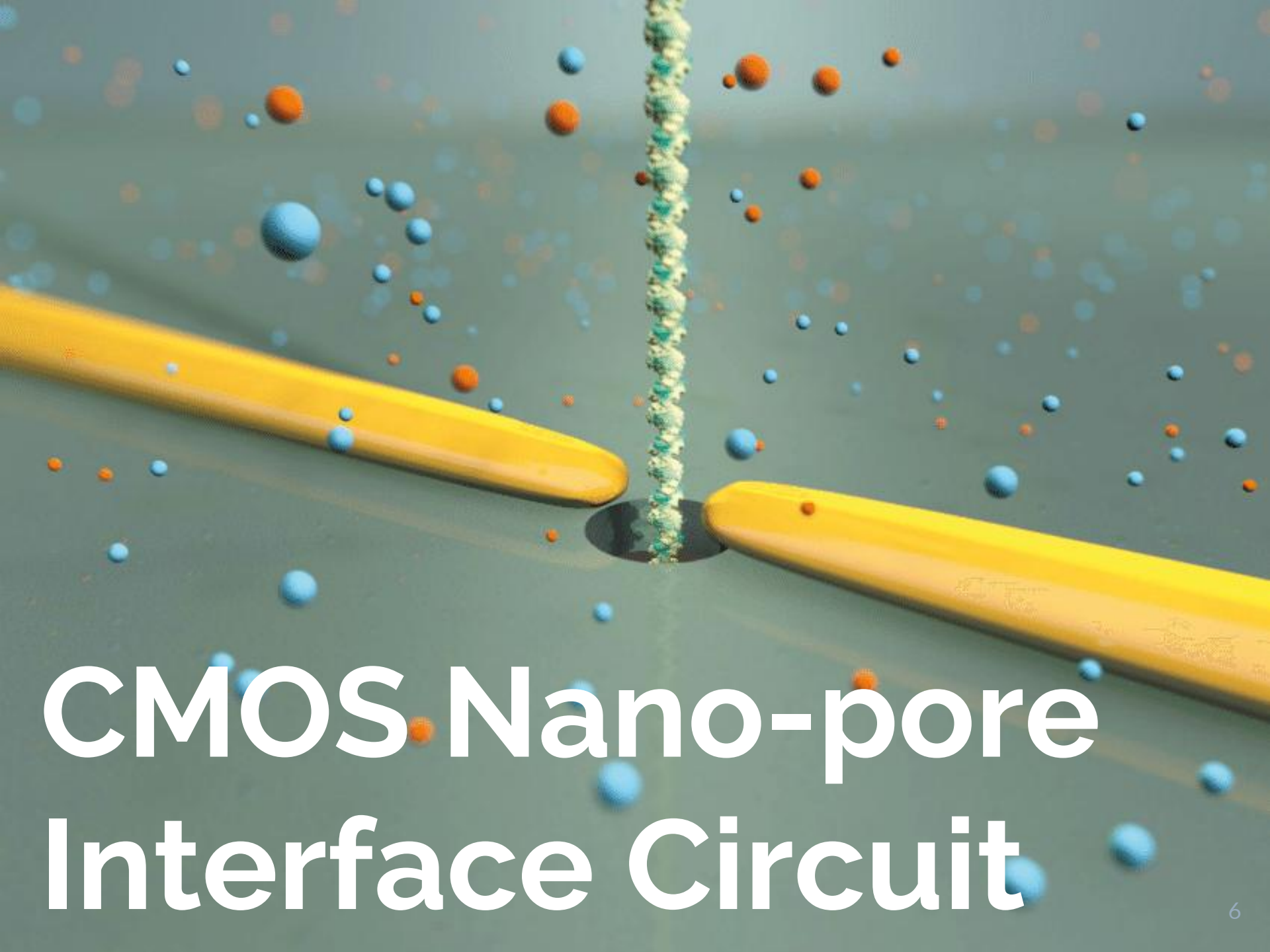
Analog Design Engineer at Texas Instruments

Project accomplishments

During my involvement with Texas Instruments I was Involved with Analog/Mixed-signal, Layout, statistical analysis, Verification, Packaging definition, and Post-silicon verification of Several LS and Half-bridge Gate Drivers

- **UCC278xx and Unreleased parts: 200V Half-Bridge driver with shutdown capabilities**
 - Lead several block designs and helped senior leadership with top-design and ESD selection
 - Acted as Layout lead and helped perform post-layout verification sims
 - Created novel solutions for safe chip-wide shutdown and enabling mechanisms to operate under external EMI and internal extreme noisy operations
- **UCC571xx: TI's first gen protection low-side Drivers with DESAT and OCP**
 - I was the lead engineer in charge of the new generation's protection features such as DESAT and OCP
 - I was the point-person for the top Aging/Reliability sims, functional testing, and initial bench validation
- **UCC274xx and UCC276xx : TI's first high-power low-side Drivers**
 - I was the in charge of responding to customer returns and troubleshooting
 - I contributed to fixes and performance optimization for the new generation of the device

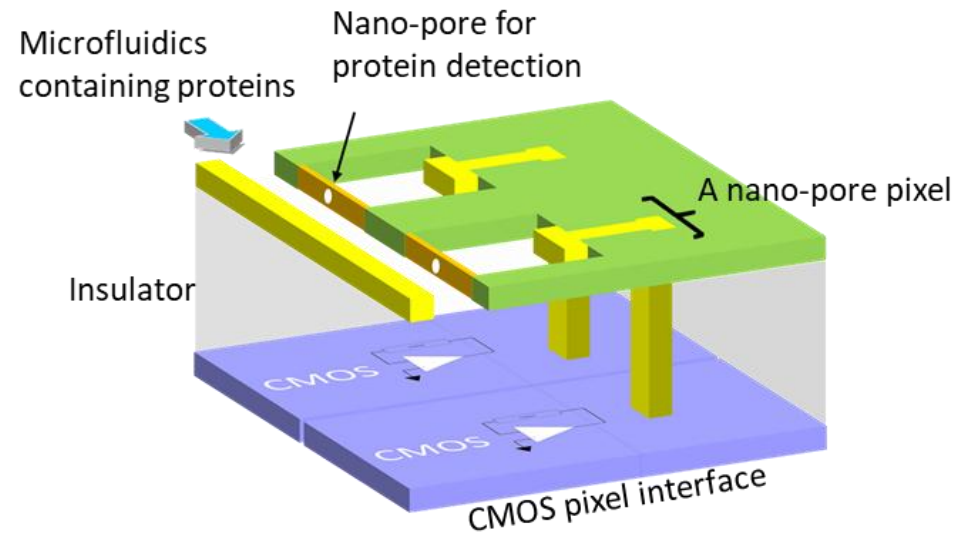




CMOS Nano-pore Interface Circuit

CMOS Nano-pore Interface

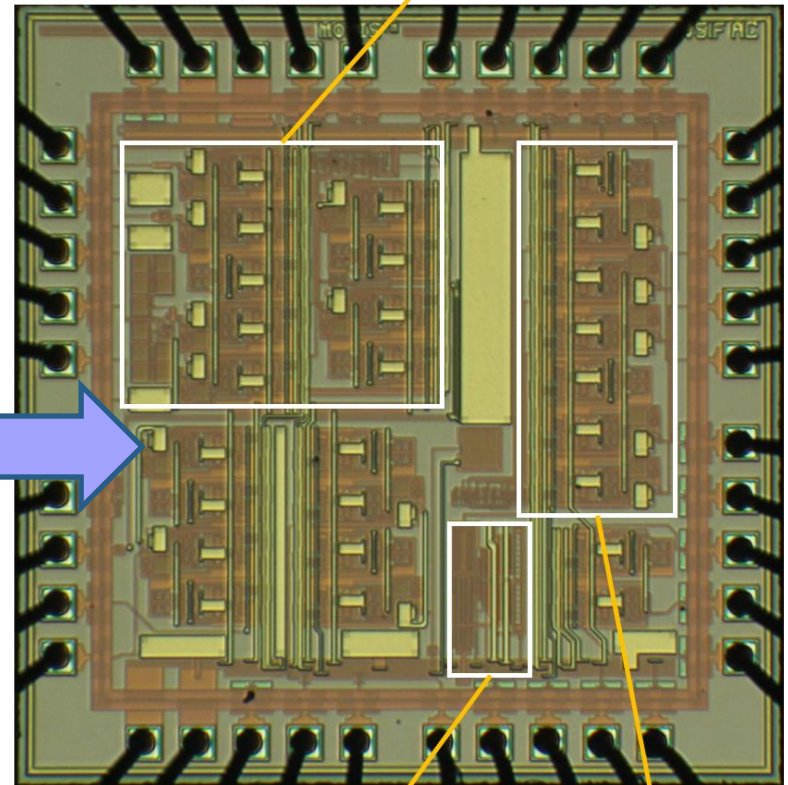
The goal is to create an interface circuit that has sufficient performance over a bandwidth of 100KHz that would allow detection of proteins passing through Nano-pores in an extremely noisy environment



My solution

I designed, Fabricated and tested a CMOS pixel interface circuit towards a lab-on-CMOS solution as shown above

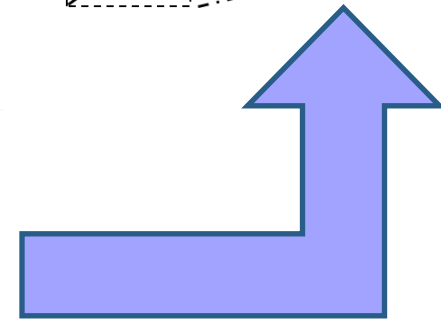
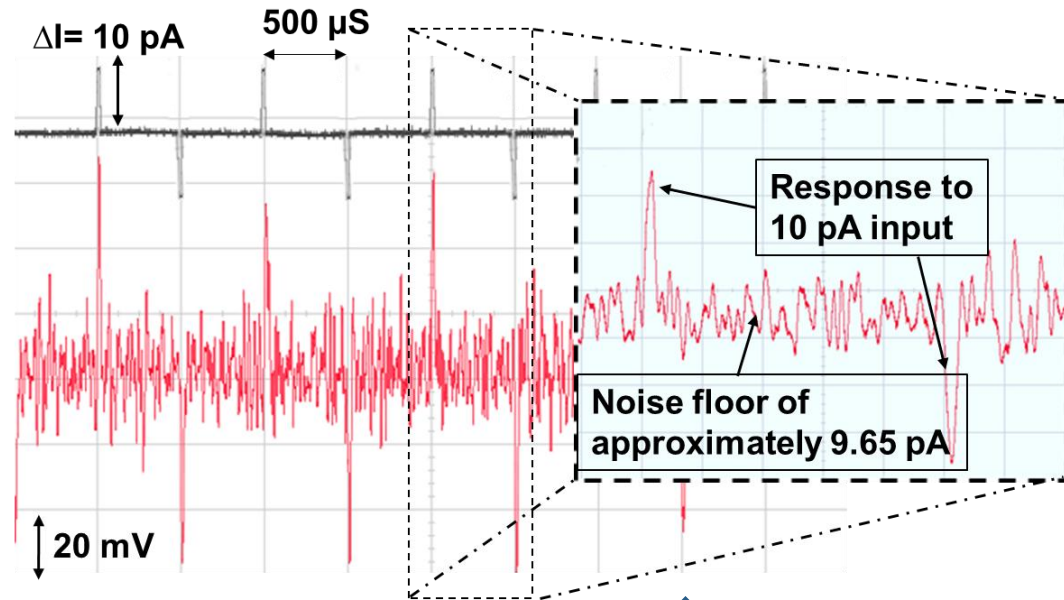
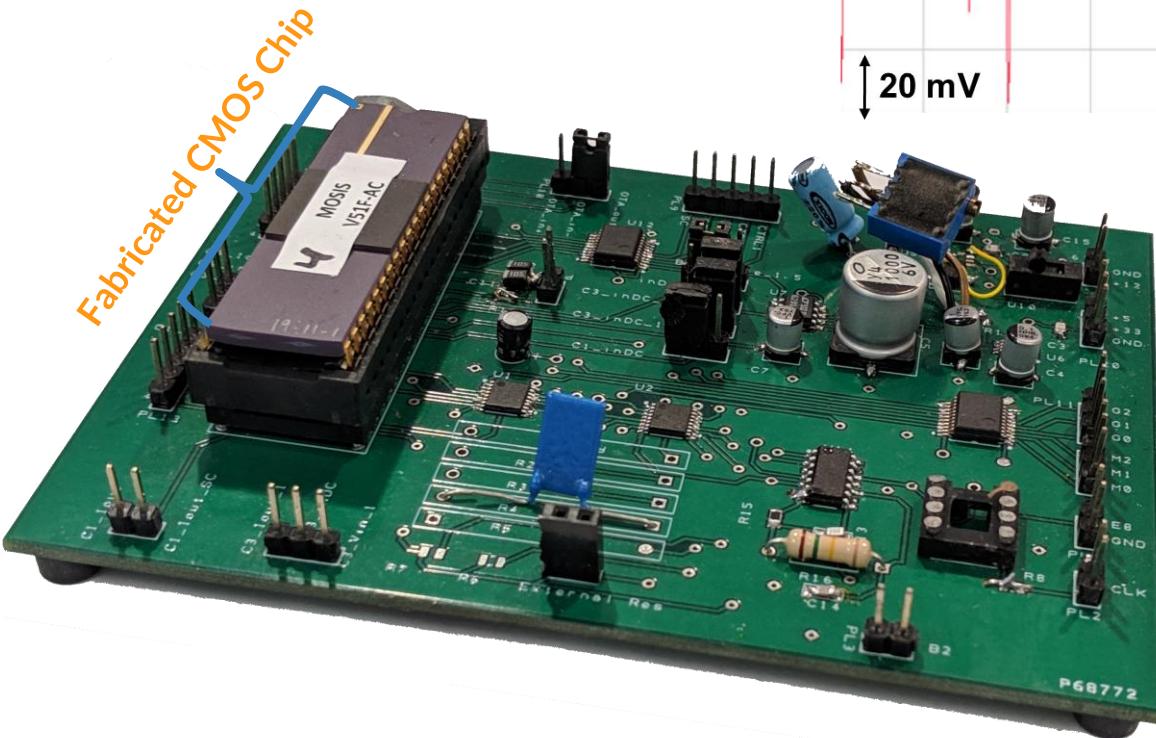
Current amplifier stage with gain of 1000 and capacitive input with accumulative noise floor of 3pA over 0-100 kHz



Bias stage

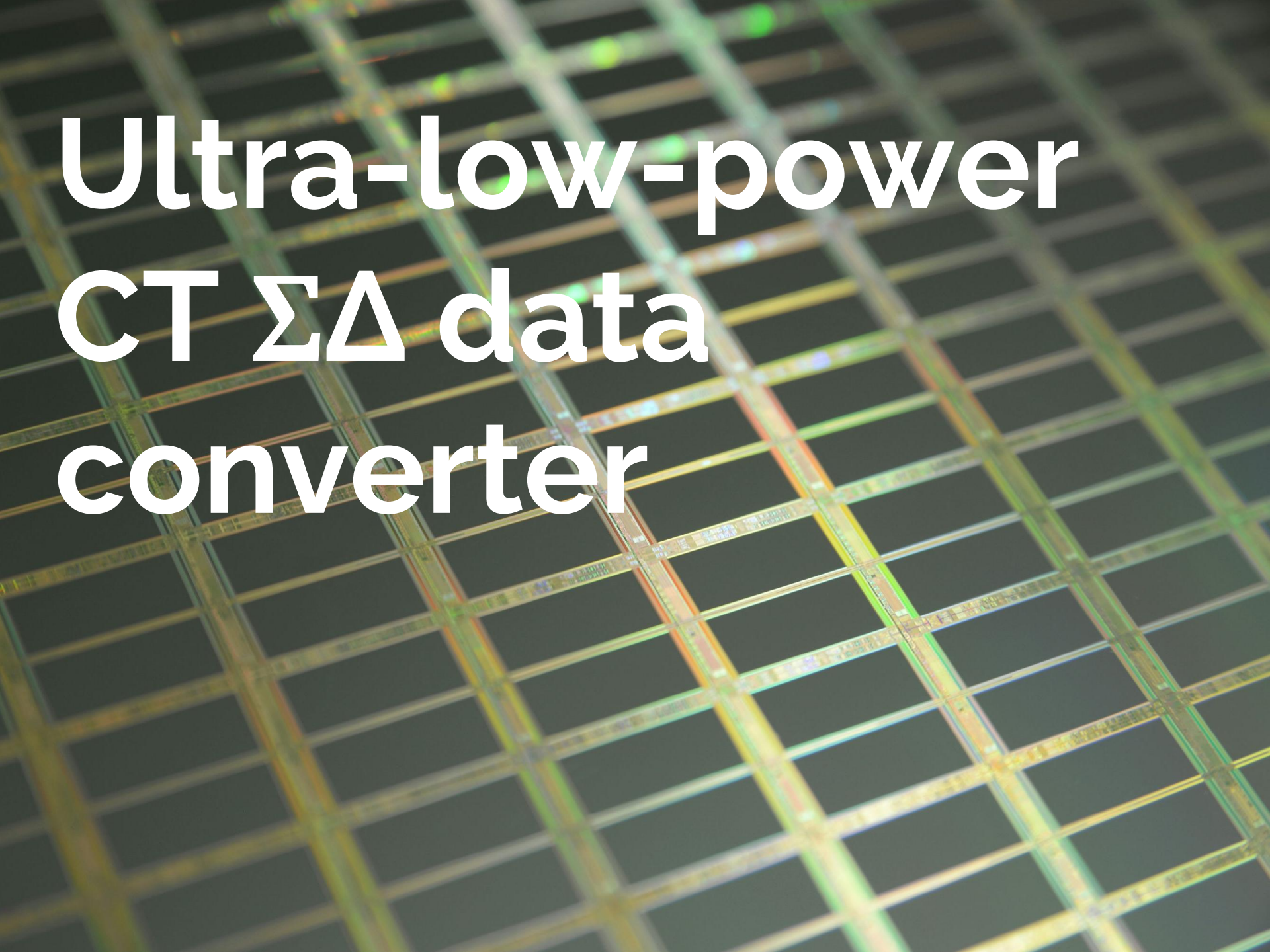
Auxiliary buffer

Test and evaluation setup

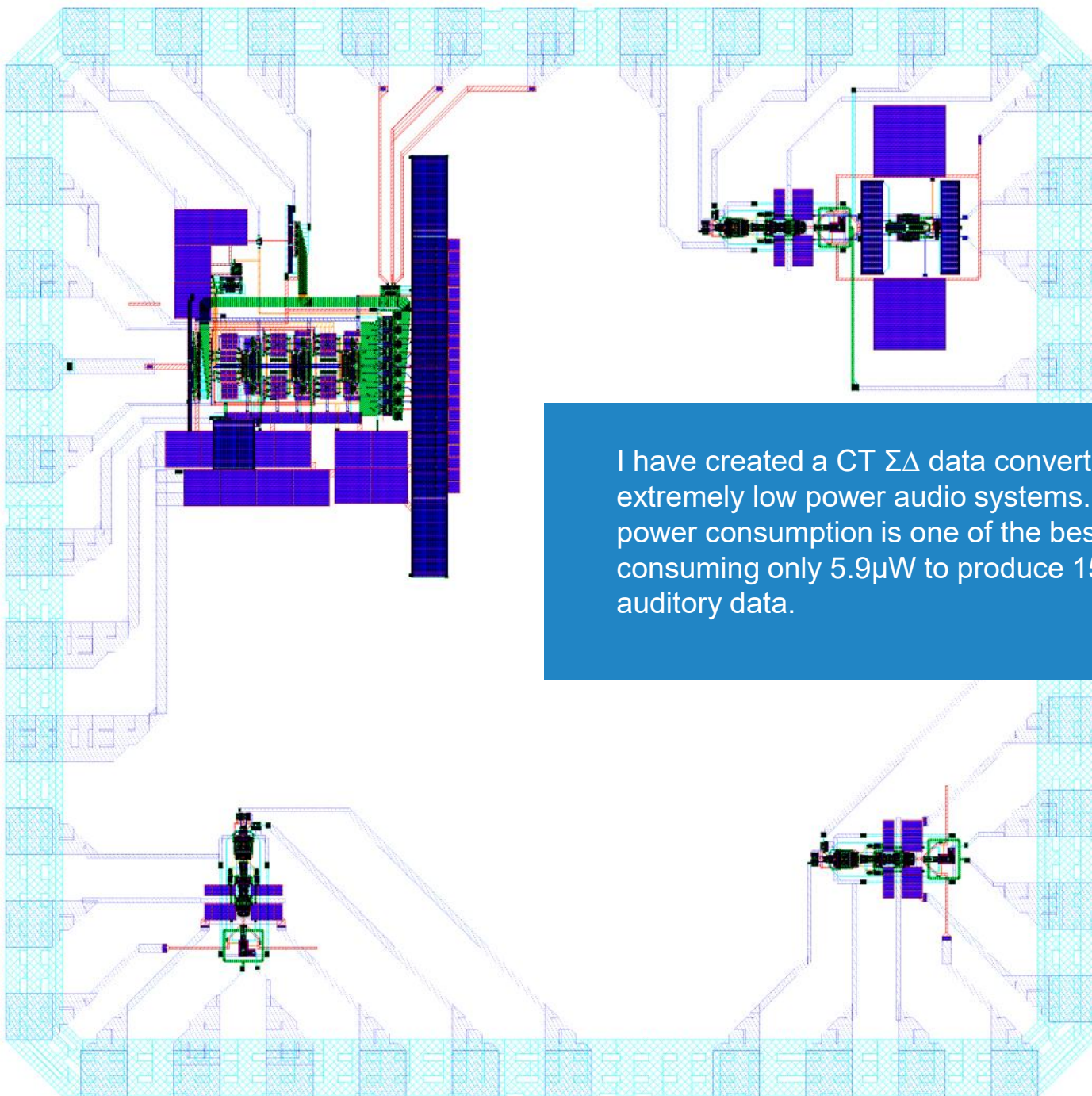


Test results

The CMOS circuit displayed exemplary performance detecting low amplitude current inputs in a very noisy environment

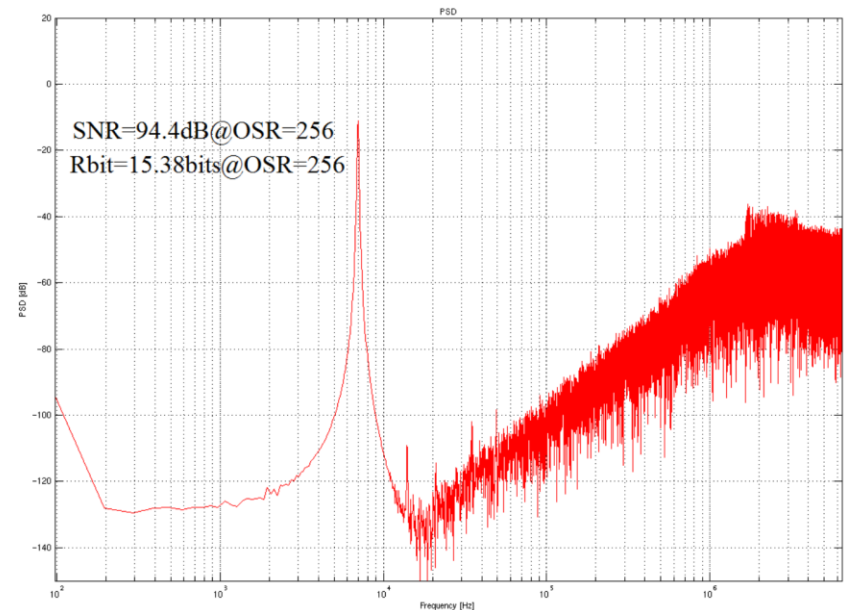
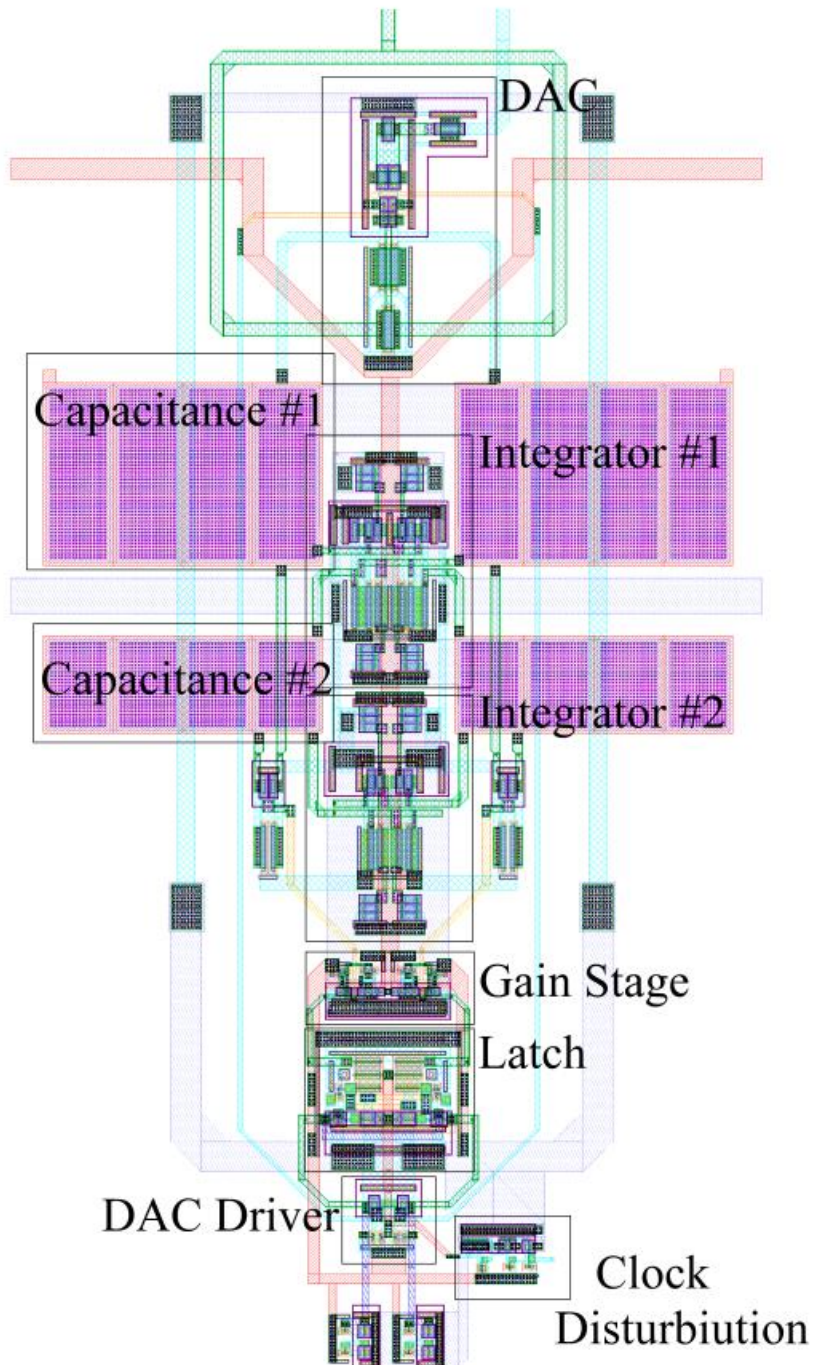
A microscopic view of a silicon chip, showing a grid of metal lines and various colored regions (green, yellow, red) representing different materials or structures. The text is overlaid on the top left of the image.

Ultra-low-power CT $\Sigma\Delta$ data converter



I have created a CT $\Sigma\Delta$ data converter for extremely low power audio systems. My design's power consumption is one of the best in its class, consuming only 5.9 μ W to produce 15 bits of auditory data.

I have created a CT $\Sigma\Delta$ data converter for extremely low power audio systems. My design's power consumption is one of the best in its class, consuming only 5.9 μ W to produce 15 bits of auditory data.



Test results

The data converter shows exemplary performance in our frequency bands of interest.

How can we facilitate
neural stimulation to
facilitate machine to
human communication?

Peripheral Nerve Stimulation

My Approach

- The goal of this project was to bridge the gap between neural stimulation techniques and human-computer interaction (HCI)
- This work aimed to transfer information to the brain using specific neural stimulation patterns called electrotactile icons
- A prototype test infrastructure was developed to activate nerve endings on the skin surface through electrotactile stimulation according to programmed patterns, enabling systematic recording of participant responses to the stimuli.



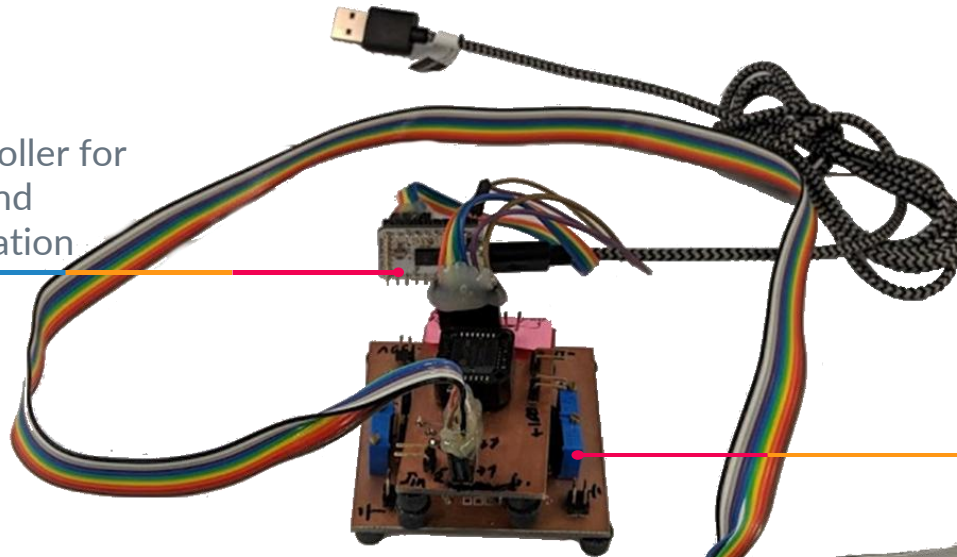
List of publications Associated with this project:

- S. Parsnejad, E. Ashoori, S. Davila-Montero, and A. J. Mason, "Distinguishable Multi-layer Frequency-modulated Waveforms for Peripheral Nerve Communication in Human Augmentation Applications," IEEE Trans. Haptics, 2026.
- S. Parsnejad, J. Brascamp, G. Pelled, and A. Mason, "A review of electrotactile stimulation for machine-to-human communication," IEEE Trans. Biomed. Eng., pp. 1–21, 2026, doi: 10.1109/TBME.2026.3661416.
- S. Parsnejad and A. J. Mason, "Etacteme Machine-to-Human Communication: A Linguistically-Inspired Experimentally-Derived Electrotactile Icon Generation Scheme," npj Biomedical Innovations (in review), pp. 1–11, 2026.
- S. Parsnejad, Enabling real-time communication for human augmentation systems via unobtrusive high bandwidth machine to human electrotactile peripheral nerve stimulation. 2022

My prototype electrotactile stimulator

Microcontroller

Local microcontroller for system control and waveform generation

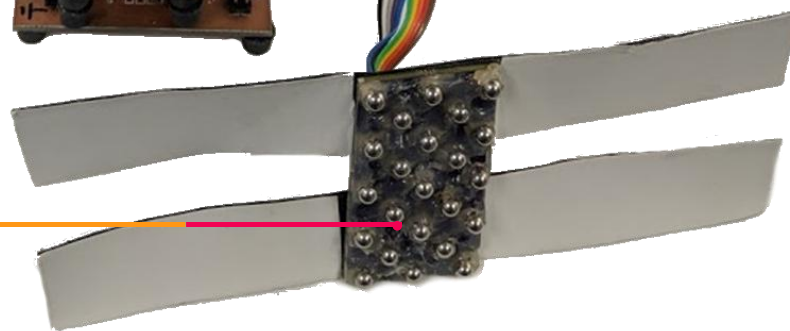


Electrotactile Charge generator

Custom made electrotactile charge generator

Electrodes

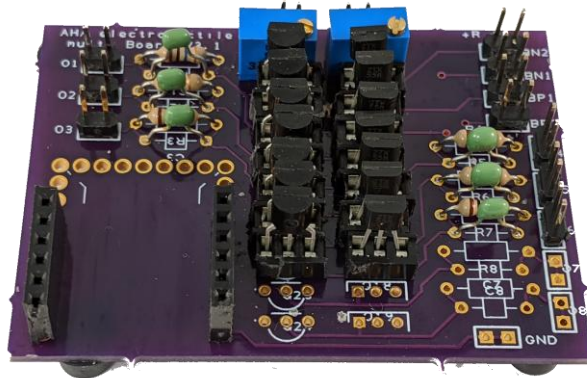
Medical-grade stainless steel electrodes for skin contact



Design Showcase

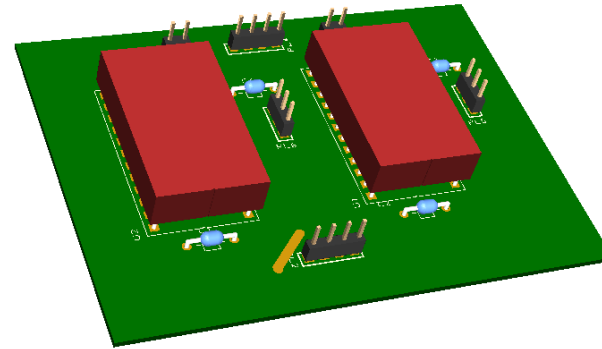
Supply generator

Very high voltage supply generator ($\pm 100\text{V}$) to facilitate electrotactile stimulations protocols



Electrodes

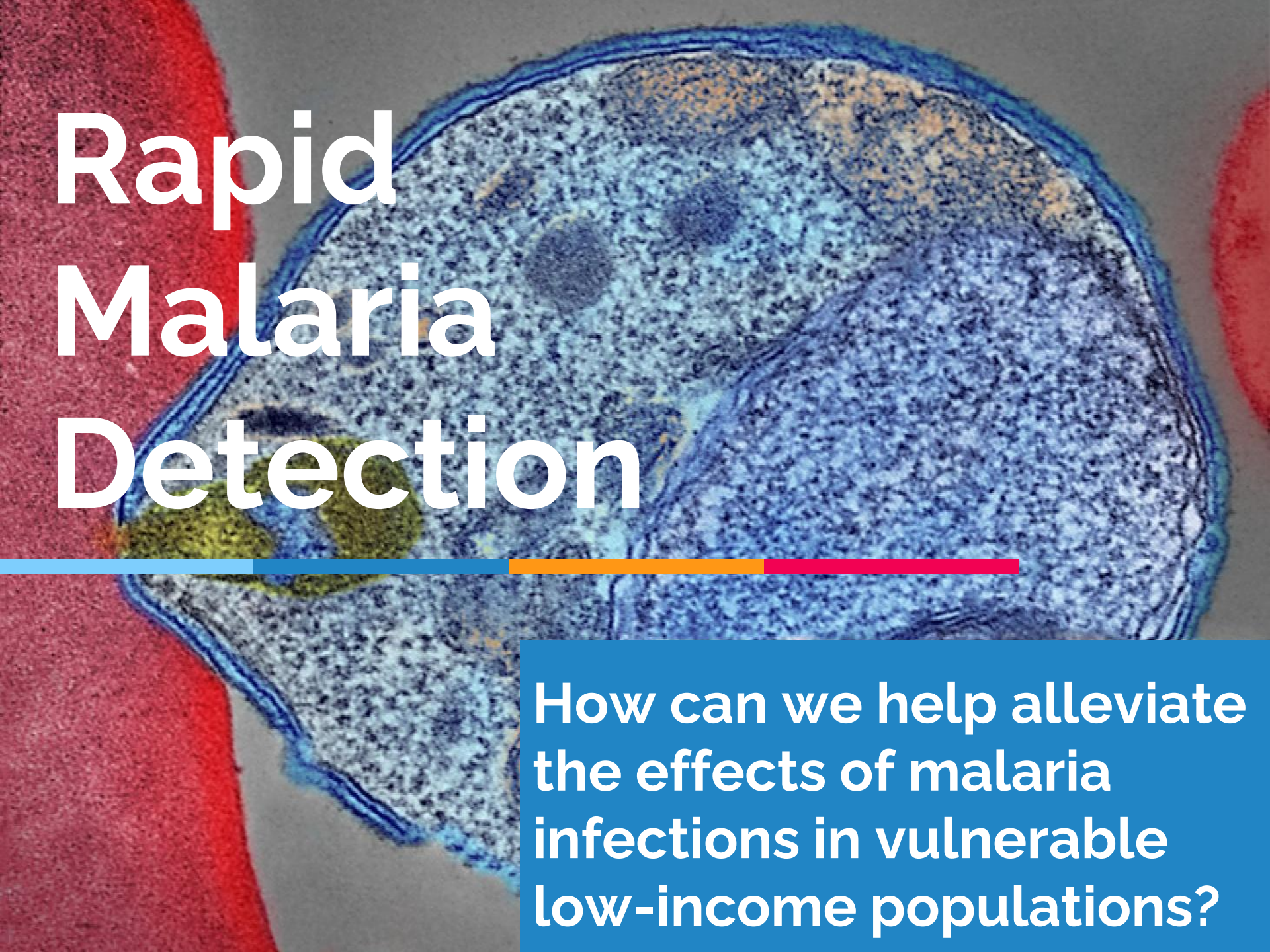
Medical-grade stainless steel electrodes for skin contact



Electrotactile Charge generator

Custom made electrotactile charge generator





Rapid Malaria Detection

How can we help alleviate the effects of malaria infections in vulnerable low-income populations?

Our solution:

aMEASURE: wearable/portable electrochemical instrument



aMEASURE is a portable/wearable electrochemical instrument that can
Adapt itself to any electrochemical sensor it is connected to
Dynamic Input current range of 115dB
Multichannel readout allowing for comparative studies

Rapid malaria detection

The goal is to leverage the flexibility and small footprint of aMEASURE to detect malaria anti bodies in blood samples. This will be done through a malaria detection chip made by my collaborators

aMEASURE

Powered by an android phone, aMEASURE (encased) will perform electrochemical tests on-the-go

Malaria detection chip

With the help of aMEASURE, this microfluidic chip made by my collaborators will detect Malaria anti-bodies



Field tests:

Our solution is
undergoing human trials
in Malawi, Africa this
Malaria season



Generation 1

First-generation prototype was constructed as a proof-of-concept for our vision for aMEASURE

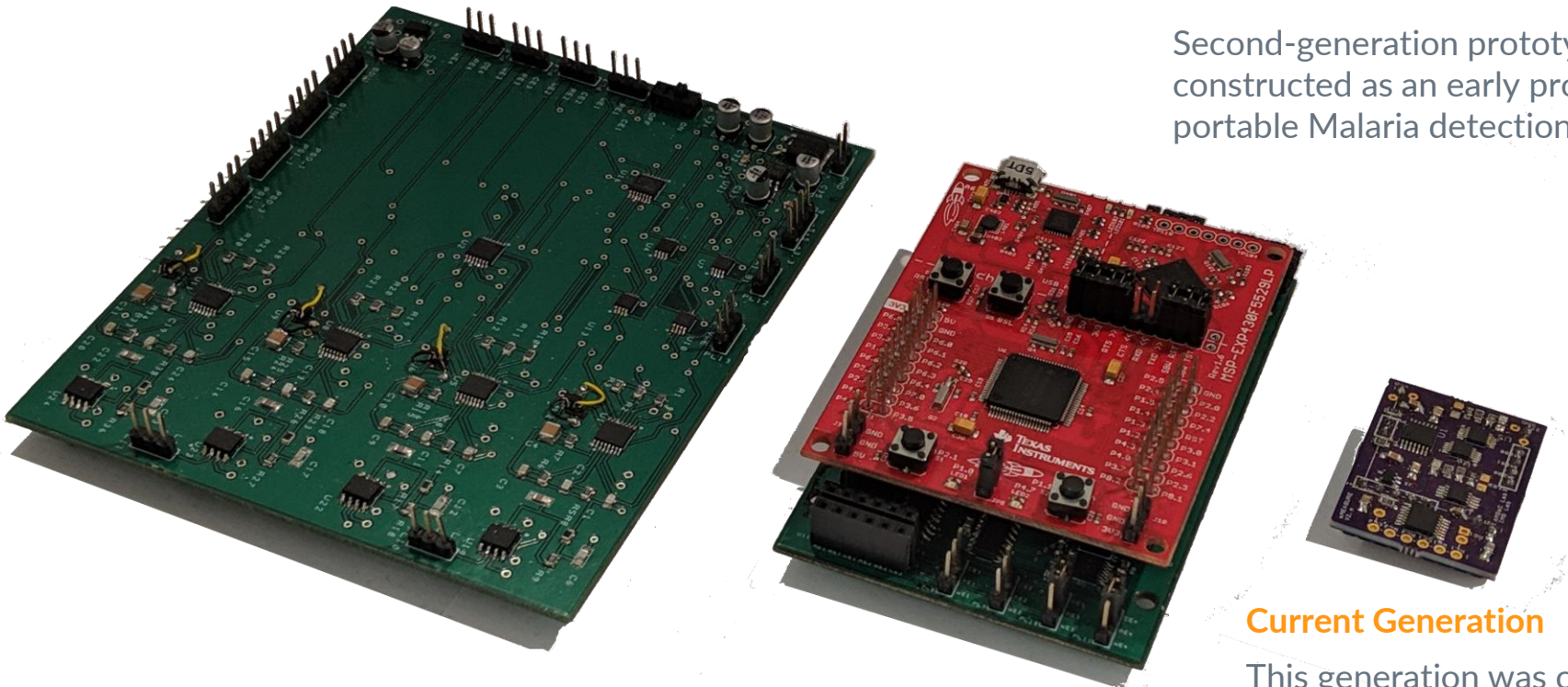
Design History

Generation 2

Second-generation prototype was constructed as an early prototype for portable Malaria detection

Current Generation

This generation was created for field tests in Africa and wearable electrochemical sensors



After identifying the problems with the current state of the art, multiple design iterations were created to test different ideas.

Thank you for viewing

Sina Parsnejad

s.parsnejad@gmail.com

Github: <https://github.com/sparsnejad/>

Resume: https://github.com/sparsnejad/Sina_Parsnejad_portfolio/blob/main/Parsnejad_Resume.pdf



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My Latest
Resume



My GitHub